

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CCT) (CBCS) Degree Examinations

December 2024 – January 2025

22CCT102 – CONSTRUCTION ECONOMICS & FINANCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

Marks BT* CO* PO*

1. a) Consider the following data of a company for the year 2023:
 Sales = Rs. 1,20,000
 Fixed cost = Rs. 25,000 Variable cost = Rs. 45,000
 Find the following:
 i) Contribution ii) Profit
 iii) BEP iv) M.S. 10 L*3 1 1
- b) The cost for the tank design as originally submitted was estimated to be Rs. 9,00,000. What are the optimum tank dimensions if the volume remains the same as for the original design? What total savings may be expected through the redesign? 10 L2 1 1
2. a) Explain different Elements of costs along with corresponding formule. 10 L2 1 1

Unit – II

3. a) Describe various types of classification of capital budgeting according to their risk. 10 L2 2 2
- b) Mention Objectives of Capital Budgeting. 10 L2 2 2
4. a) In a particular locality of a state, the vehicle users take a roundabout route to reach certain places because of the presence of a river. This results in excessive travel time and increased fuel cost. So, the state government is planning to construct a bridge across the river. The estimated initial investment for constructing the bridge is Rs. 40,00,000. The estimated life of the bridge is 15 years. The annual operation and maintenance cost is Rs. 1,50,000. The value of fuel savings due to the construction of the bridge is Rs. 6,00,000 in the first year and it increases by Rs. 50,000 every year thereafter till the end of the life of the bridge. Check whether the project is justified based on BC ratio by assuming an interest rate of 12%, compounded annually. 2 L4 2 2

Unit – III

5. a) A firm is considering replacement of an equipment, whose first cost is Rs. 4,000 and the scrap value is negligible at the end of any year. Based on experience, it was found that the maintenance cost is zero during the first year and it increases by Rs. 200 every year thereafter.
 (a) When should the equipment be replaced if $i = 0\%$?
 (b) When should the equipment be replaced if $i = 12\%$? 20 L4 3 3
6. a) A steel highway bridge must either be reinforced or replaced. Reinforcement would cost Rs. 6,60,000 and would make the bridge fit for an additional five years of service. If it is reinforced, it is estimated that its net salvage value would be Rs. 4,00,000 at the time it is retired from service. The new prestressed concrete bridge would cost Rs. 15,00,000 and would meet the foreseeable requirements of the next 40 years. Such a bridge would have no salvage value. It is estimated that the annual maintenance cost of the reinforced bridge would exceed that of the concrete bridge by Rs. 96,000. If the bridge is replaced by a new prestressed concrete bridge, the scrap value of the steel would exceed the demolition cost by Rs. 4,20,000. Assume that the money costs the state 10%. What would you recommend? 20 L4 3 3

Unit – IV

7. a) From the following transactions find out the nature of account and also state which account should be debited and which account should be credited:

- i. Salary paid
- ii. Interest received
- iii. Machinery purchased for cash
- iv. Building sold
- v. Outstanding salary
- vi. Received cash from Ramesh
- vii. Proprietor introduced capital
- viii. Dividend received
- ix. Commission paid

Furniture purchased for cash

20 L4 4 4

8. a) Journalize the following transactions in the books of Nitte Ltd. 2023 March

Date	Particulars	Rs. in lakhs
1	Started business with cash	4,500
1	Paid into bank	2,500
2	Goods purchased for cash	1,500
3	Purchase of furniture and payment by cheque	500
5	Sold goods for cash	600
8	Sold goods to Rosy	400
10	Goods purchased from Thomas	700
12	Goods returned to Thomas	100
15	Sold goods to Rahavan for cash	250
18	Cash received from Rosy Rs. 396 & discount allowed to her Rs. 4	
21	Withdraw from bank for private use	100
21	Withdraw from bank for use in the business	500
25	Paid telephone rent for one year	40
28	Cash paid to Rosy in full settlement of her account	594
30	Paid for Stationary	20
	Rent Paid	100
	Salaries to Staff	250

20 L4 4 4

Unit – V

9. a) From the following information of Nitte Engineers Ltd. On 31st March, 2023 you are required to prepare the Trading, Profit and Loss A/c and Balance sheet.

Particulars	Rs.	Particulars	Rs.
Opening stock	5,000	Capital	89,500
Bills Receivable	22,500	Commission (Cr.)	2,000
Purchases	1,95,000	Return Outward	2,500
Wages	14,000	Trade Expenses	1,000
Insurance	5,500	Office Fixtures	5,000
Sundry Debtors	1,50,000	Cash in Hand	2,500
Carriage Inward	4,000	Cash at Bank	23,750
Commission (Dr.)	4,000	Rent & Rates	5,500
Interest on Capital	3,500	Carriage outward	7,250
Stationery	2,250	Sales	2,50,000
Return Inward	6,500	Bills Payable	15,000
		Creditors	98,250
		Closing Stock	12,500

10. a) Explain IFRS and its Requirements.
 b) Differentiate between Profit and Loss Account and Balance Sheet.

20	L4	5	5
10	L2	5	5
10	L2	5	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
